Suppose you decide to organize a Snakes and Ladders competition with n participants. In this competition, each game of Snakes and Ladders involves three players. After the game is finished, they are ranked first, second, and third. Each player may be involved in any (non-negative) number of games, and the number need not be equal among players.

At the end of the competition, m games have been played. You realize that you forgot to implement a proper rating system, and therefore decide to produce the overall ranking of all n players as you see fit. However, to avoid being too suspicious, if player A ranked better than player B in at least one game, then A must rank better than B in the overall ranking.

You are given the list of players and their rankings in each of the m games. Describe and analyze an algorithm that produces an overall ranking of the n players that is consistent with the individual game rankings, or correctly reports that no such ranking exists.

1. It is always good practice to come up with an example to make sure you understand the problem statement. Give an example competition with four distinct players and the outcome of three games. Propose an overall ranking of the players. Is it valid?

2. Your algorithm must be a reduction to a graph problem. Draw the graph you would build for your instance and use it to demonstrate that your overall ranking is or is not valid.

3. You are now ready to write down your full algorithm. You can do so by answering the following questions:

- What are the original parameters of the problem? (That is, the variables describing the input?) You will need this to analyze the runtime of your algorithm.

- What are the vertices? What does each vertex represent?

- What are the edges? Are they directed or undirected?

- If the vertices and/or edges have associated values, what are they?

- **What is the precise relationship between your graph and the stated problem?**

- What standard graph algorithm are you using to solve the problem?

- What is the running time of your entire algorithm, *including* the time to build the graph, *as a function of the original input parameters*?